Concept design

DE-BUG
GANG

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Two Tools, One System

One app for staff at their desk. One app for people in the field. Both talk to each other.

Staff Dashboard

For staff • On a computer • Data & reports

- See live health stats for the arboretum
- Explore the map — green vs. paved areas
- Auto-generate sustainability reports
- Add or upload data safely

synced

Field App

For visitors & staff • On your phone • Works offline

- Look up plants and animals on the spot
- Scan QR codes on trees to learn more
- Submit what you see in seconds
- Browse the arboretum map by zone

What you log in the field shows up straight away in the staff dashboard, no manual transfer needed.

Public Visitor - Mobile App

Interactive Map

- Clickable elements
- Covering landscape areas first, species later

Location

- Marks coordinates per upload

Upload data

- in app camera button

Interactive elements

- contribution based leaderboards, top 5 gets badge
- Public progression coverage of how many explored

HOoS - Desktop

Dashboard

- Live updating health statistics
- Permission system
- Historic view of data
- Percentage of hardscape vs softscape

Generate Report

- Export PDF / raw data

Data

Data Reliability Tiers:

1. Sensor values
2. Arboretum Staff Data
3. Visitor + pictures
4. Vistor no pictures

Sensor Data to collect:

- Soil
- Humidity
- Light

AI Validation

Species Image Classification

Anomaly detection

- Green Space Anomaly Detection
- Species Anomaly Detection

Inacuracy notification

- Notify user in mobile app if AI decides to correct anomaly data entry

Tree Health Prediction

- Sensor data